

Primes and Geodesics

Selberg's Trace Formula and Applications

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Motivation: A Curious Coincidence

We have two identical asymptotic equations:

$$\pi(x) \sim \frac{x}{\log x} \longleftrightarrow \Pi(x) \sim \frac{x}{\log x}$$

But the functions are *totally different*. The prime counting function $\pi(x)$ is the number of primes less than x .

The function $\Pi(x)$ counts the number of *primitive closed geodesics* of length less than $\log x$, inside a compact hyperbolic surface.

This is a consequence of *Selberg's Trace Formula*.

- 1 Selberg's Trace Formula for compact hyperbolic surfaces
- 2 Sketch of proof (LHS & RHS)
- 3 Application: Prime Geodesic Theorem
- 4 Further questions (including my research)

> Setup: *compact hyperbolic surfaces*.

> Identified with quotients $\Gamma \backslash \mathbb{H}^2$ by a discrete, nontorsion, cocompact subgroup $\Gamma \subset \mathrm{PSL}_2(\mathbb{R}) = \mathrm{Isom}^+(\mathbb{H}^2)$.

> Model: $\mathbb{H}^2 = \{x + iy \in \mathbb{C} \mid y > 0\}$ with $ds^2 = y^{-2}(dx^2 + dy^2)$ and $\Delta = y^2(\partial_x^2 + \partial_y^2)$. View functions on $\Gamma \backslash \mathbb{H}^2$ as Γ -periodic functions on $\mathbb{H}^2$. Integrate on a fundamental domain of Γ in $\mathbb{H}^2$.

Solving $(\Delta + \lambda)f = 0$, we get $0 = \lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots \rightarrow \infty$ and $f = \varphi_j$. Furthermore, we often use the parametrization $\lambda_j = \frac{1}{4} + \rho_j^2$. (The sign of ρ_j won't matter.)

Selberg's Trace Formula

Theorem (Trace formula)

For sufficiently nice Fourier pairs h and g , we have

$$\sum_{j=0}^{\infty} h(\rho_j) = \frac{\text{Area}(\Gamma \backslash \mathbb{H}^2)}{4\pi} \int_{-\infty}^{\infty} h(\rho) \tanh(\pi\rho) \rho \, d\rho \\ + \sum_{p \in H_*} \sum_{n=1}^{\infty} \frac{\ell_p g(n\ell_p)}{2 \sinh(n\ell_p/2)}$$

where H_* is the set of *primitive* representatives of nontrivial conjugacy classes in Γ and ℓ_p is the length of the geodesic corresponding to p .

We will sketch a proof.

Sketch of proof

The trace formula arises by computing the trace of an integral operator

$$[L_\Gamma f](z) := \int_{\Gamma \backslash \mathbb{H}^2} K_\Gamma(z, w) f(w) d\mu(w), \quad K_\Gamma := \sum_{\gamma \in \Gamma} k(\gamma z, w)$$

in two different ways.

Definition (Selberg's Point-Pair Invariant)

The point-pair invariant $k(z, w)$ is defined via integrating $h(\rho)$ against the Green function $G_\rho(z, w)$ for $\Delta + \lambda$ on $\mathbb{H}^2$. It is important that this depends only on $\text{dist}(z, w)$.

Sketch of proof: LHS

As $\Gamma \backslash \mathbb{H}^2$ is compact, we can expand $K_\Gamma(z, w)$ as a function of z in terms of the L^2 -orthonormal basis of Laplacian eigenfunctions φ_j :

$$K_\Gamma(z, w) = \sum_{j=0}^{\infty} c_j \varphi_j(z)$$

where $c_j = \int_{\Gamma \backslash \mathbb{H}^2} K_\Gamma(z, w) \varphi_j(z) d\mu(z)$. As $k(z, w)$ is isometry-invariant, L_Γ commutes with Δ , and in fact we have $c_j = h(\rho_j) \varphi_j(w)$.

> Substituting c_j back, letting $z = w$ and integrating over $\Gamma \backslash \mathbb{H}^2$ gives

$$\int_{\Gamma \backslash \mathbb{H}^2} K_\Gamma(z, z) d\mu(z) = \sum_{j=0}^{\infty} h(\rho_j) \int_{\Gamma \backslash \mathbb{H}^2} \varphi_j(z)^2 d\mu(z) = \sum_{j=0}^{\infty} h(\rho_j).$$

This proves the LHS.

Sketch of proof: Geometric Side (RHS)

Start with writing

$$K_{\Gamma}(z, z) = \overset{\text{easy}}{k(z, z)} + [\overset{\text{hard}}{K_{\Gamma}(z, z) - k(z, z)}].$$

> Main idea: reduce the hard integral

$$\int_{\Gamma \backslash \mathbb{H}^2} [K_{\Gamma}(z, z) - k(z, z)] d\mu(z)$$

to the case where Γ is a cyclic group.

Sketch of proof: Geometric Side (RHS)

If $\Gamma = \langle \gamma \rangle$, its fundamental domain has a simple form

$$\text{Fund}(\Gamma) \cong \{s \mid 1 \leq \text{Re}(s) \leq e^{\ell_\gamma}\}$$

where $\ell_\gamma = 2 \cosh^{-1}(|\text{tr}(\gamma)|/2)$.

Proposition (A)

If Γ and ℓ_γ are as above,

$$\int_{\Gamma \backslash \mathbb{H}^2} [K_\Gamma(z, z) - k(z, z)] d\mu(z) = \sum_{n=1}^{\infty} \frac{\ell_\gamma g(n\ell_\gamma)}{\sinh(n\ell_\gamma/2)}$$

where g is the Fourier transform of h .

Sketch of proof: Geometric Side (RHS)

We can rewrite

$$K_{\Gamma}(z, z) - k(z, z) = \sum_{e \neq \gamma \in \Gamma} k(\gamma z, z) = \sum_C \sum_{\gamma \in C} k(\gamma z, z)$$

where C runs through nontrivial conjugacy classes in Γ . Furthermore,

$$\sum_{\gamma \in H} \sum_{g \in Z_{\gamma} \backslash \Gamma} k(g^{-1} \gamma g z, z) \quad \because \text{orbit-stabilizer}$$

where H is a set of representatives for C , Z_{γ} is the centralizer of γ (which is the stabilizer of conjugation) and g is a representative of a coset of Z_{γ} .

Sketch of proof: Geometric Side (RHS)

Hence we are left with

$$\sum_{\gamma \in H} \sum_{g \in Z_\gamma \setminus \Gamma} k(g^{-1}\gamma gz, z) = \sum_{\gamma \in H} \sum_{g \in Z_\gamma \setminus \Gamma} k(\gamma gz, gz) \quad \because \text{isom-invariant}$$

It is a fact¹ that every Z_γ is cyclic with generators called *primitive elements*, hence letting $H_* \subset H$ be the subset of primitive representatives we may write

$$\sum_{p \in H_*} \sum_{g \in Z_p \setminus \Gamma} \sum_{n=1}^{\infty} k(p^n gz, gz) = \frac{1}{2} \sum_{p \in H_*} \sum_{g \in Z_p \setminus \Gamma} [K_{Z_p}(gz, gz) - k(gz, gz)],$$

omitting details.

¹This is nontrivial and due to the classification of isometries

Sketch of proof: Geometric Side (RHS)

As the $g \cdot \text{Fund}(\Gamma)$ glue over $g \in Z_p \setminus \Gamma$ to make $\text{Fund}(Z_p)$, integrating over $\Gamma \setminus \mathbb{H}^2$ gives

$$\frac{1}{2} \sum_{p \in H_*} \int_{\text{Fund}(Z_p)} [K_{Z_p}(z, z) - k(z, z)] .$$

As $Z_p = \langle p \rangle$, by Proposition (A) this is

$$\frac{1}{2} \sum_{p \in H_*} \sum_{n=1}^{\infty} \frac{\ell_p g(n\ell_p)}{\sinh(n\ell_p/2)} .$$

Sketch of proof: Conclusion

The 'easy' part is just

$$k(z, z) = \frac{1}{4\pi} \int_{-\infty}^{\infty} h(\rho) \tanh(\pi\rho) \rho \, d\rho,$$

so integrating over $\Gamma \backslash \mathbb{H}^2$ is just multiplication by $\text{Area}(\Gamma \backslash \mathbb{H}^2)$. Therefore we have

$$\sum_{j=0}^{\infty} h(\rho_j) = \frac{\text{Area}(\Gamma \backslash \mathbb{H}^2)}{4\pi} \int_{-\infty}^{\infty} h(\rho) \tanh(\pi\rho) \rho \, d\rho + \sum_{p \in H_*} \sum_{n=1}^{\infty} \frac{\ell_p g(n\ell_p)}{2 \sinh(n\ell_p/2)}.$$

This concludes the proof of Selberg's Trace Formula.

Application: Prime geodesic theorem

By choosing smooth approximates $g_{x,\epsilon}$ of a certain cutoff function, and applying Selberg Trace to $g_{x,\epsilon}$ and $h_{x,\epsilon}$, we may write

$$\sum_{j=0}^{\infty} h_{x,\epsilon}(\rho_j) = \frac{\text{Area}(\Gamma \backslash \mathbb{H}^2)}{4\pi} \int_{\mathbb{R}} h_{x,\epsilon}(r) \tanh(\pi r) r \, dr + \Psi_{\epsilon}(x)$$

where the term defined over primitive representatives is written $\Psi_{\epsilon}(x)$.

By direct computation of the other terms one can show $\Psi_{\epsilon}(x) \sim x$, which implies $\Psi(x) \sim x$ where $\Psi(x)$ is the (scaled) *Chebyshev function* for the geodesic counting function $\Pi(x)$, just as in the proof of PNT. This implies the Prime Geodesic Theorem.

See §9.6 of [3] for details.

Further Questions and Generalizations

Some questions:

- 1 Does the Trace Formula generalize to other compact manifolds, for example of nonconstant curvature?
- 2 Does the Prime Geodesic Theorem generalize to “prime geodesic surfaces” in $\Gamma \backslash \mathbb{H}^3$?

Further Questions and Generalizations

Question

Does the Trace Formula generalize to other compact manifolds, for example of nonconstant curvature?

For most, no. Recall that the Green function G_ρ on $\mathbb{H}^2$ depended only on $\text{dist}(z, w)$. For this to happen, the isometry group must act transitively on pairs of points with same distance; these spaces are very uniform:

- ① rank 1 symmetric (eg. $\mathbb{S}^n, \mathbb{H}^n, \dots$),
- ② Euclidean ($\mathbb{R}^n$).

But for these spaces it does generalize, and $\mathbb{R}^n/\mathbb{Z}^n$ is just Poisson summation.

Further Questions and Generalizations

Question

Does the Prime Geodesic Theorem generalize to “prime geodesic surfaces” in $\Gamma \backslash \mathbb{H}^3$?

For certain Γ , yes. More precisely, when $\Gamma_d := \mathrm{PSL}_2(\mathcal{O}_K) \subset \mathrm{PSL}_2(\mathbb{C})$ for $K = \mathbb{Q}(\sqrt{-d})$, we have $\Pi(x) \sim Cx$ for a constant C . Finding C follows from clasifying the maximal Fuchsian subgroups of Γ_d , which was done for

- ① $d = 1$ by Maclachlan and Reid (1991),
- ② $d \equiv 3 \pmod{4}$ such that $4 \nmid h_K$ by Jung (2019).

I did the $d = 2$ case (arXiv:2601:15700), and found

$$C = \frac{45}{16\pi} \cdot \prod_{p \text{ prime}} \left[1 - \frac{1}{p} + \frac{1}{p + \left(\frac{-2}{p}\right)} \right] \approx 0.88.$$

Thank you for listening!

References:

- ① Selberg's Trace Formula: An Introduction, J. Marklof (Main Ref.)
- ② The Selberg Trace Formula, E. Assing
- ③ Geometry and Spectra of Compact Riemann Surfaces, P. Buser